Castigliano's Theorem

Introduction

In the previous chapter, we have shown that the internal force/stress and displacement can be exposed by the discontinued variation with regional partition. We shall further show that, for a linear elastic structure that consists of uncoupled 1-D elements (space truss and/or frame), the variation of energy functional (Lagrangian) can be used to extract the real internal force/displacement at any point of the 1-D elements directly.

Please keep in mind that three basic assumptions must be satisfied in order to use Castigliano's Theorem:

1. **The governing equations are linear for all variables (linear)**
2. **Energy functional exists for the system (elastic)**
3. **All variables of the energy functional are uncoupled**

Principle of Virtual Work for Space Frame (Euler Bernoulli Beam)

Euler Bernoulli Beam

We shall use the energy functional of Euler Bernoulli Beam without derivation:

$$\frac{\partial v_\epsilon(u', v'', w'', \theta')}{\partial u'} = P$$

$$\frac{\partial v_\epsilon(u', v'', w'', \theta')}{\partial v''} = M_y \quad M_y' = V_y$$

$$\frac{\partial v_\epsilon(u', v'', w'', \theta')}{\partial w''} = M_z \quad M_z' = V_z$$

$$\frac{\partial v_\epsilon(u', v'', w'', \theta')}{\partial \theta'} = T_x$$

Principle of Virtual Displacement

$$\delta V_\epsilon(\boldsymbol{u}) = \sum_{Element} \int \left[P\delta u' + M_y \delta v'' + M_z \delta w'' + T\delta\theta'\right] ds$$

$$= \sum_{Element} \int_0^l \left[f\delta u - q_y \delta v - q_z \delta w + r\delta\theta\right] ds + \sum_{Joint,q} \boldsymbol{Q}_J \cdot \boldsymbol{\delta u}_J + \sum_{Joint,h} \boldsymbol{R}_J \cdot \boldsymbol{\delta u}_J$$

$$\delta W_{\epsilon,ext}(\boldsymbol{u}) = \sum_{Element} \int_0^l \left[f\delta u - q_y \delta v - q_z \delta w + r\delta\theta\right] ds + \sum_{Joint,q} \boldsymbol{Q}_J \cdot \boldsymbol{\delta u}_J + \sum_{Joint,h} \boldsymbol{R}_J \cdot \boldsymbol{\delta u}_J$$

$$W_{\epsilon,ext}(\boldsymbol{u}) = \sum_{Element} \int_0^l \left[fu - q_y v - q_z w + r\theta\right] ds + \sum_{Joint,q} \boldsymbol{Q}_J \cdot \boldsymbol{u}_J + \sum_{Joint,h} \boldsymbol{R}_J \cdot \boldsymbol{u}_J$$

Please note that $\boldsymbol{u}$ is the generalized displacement field, $(\boldsymbol{Q}_J, \boldsymbol{R}_J)$ are the generalized nodal force:

$$\begin{aligned}\boldsymbol{u} = (Global)&: U\boldsymbol{E}_1 + V\boldsymbol{E}_2 + W\boldsymbol{E}_3 + \Theta\boldsymbol{E}_4 + \mathrm{A}\boldsymbol{E}_5 + \mathrm{B}\boldsymbol{E}_6 \\ &= (Local): u\boldsymbol{e}_1 + v\boldsymbol{e}_2 + w\boldsymbol{e}_3 + \theta\boldsymbol{e}_4 + \alpha\boldsymbol{e}_5 + \beta\boldsymbol{e}_6\end{aligned}$$

$$\begin{aligned}\boldsymbol{Q}_J, \boldsymbol{R}_J = (Global)&: F_X\boldsymbol{E}_1 + F_Y\boldsymbol{E}_2 + F_Z\boldsymbol{E}_3 + M_X\boldsymbol{E}_4 + M_Y\boldsymbol{E}_5 + M_Z\boldsymbol{E}_6 \\ &= (Local): P\boldsymbol{e}_1 - V_y\boldsymbol{e}_2 - V_z\boldsymbol{e}_3 + T\boldsymbol{e}_4 - M_z\boldsymbol{e}_5 + M_y\boldsymbol{e}_6\end{aligned}$$

Lagrangian:

$$\mathcal{L}_\epsilon(\boldsymbol{u}) = V_\epsilon(\boldsymbol{u}) - W_{\epsilon,ext}(\boldsymbol{u})$$

$$\begin{aligned}\delta\mathcal{L}_\epsilon(\boldsymbol{u}) = &\sum_{Element} \int \left[P\delta u' + M_y\delta v'' + M_z\delta w'' + T\delta\theta'\right] ds \\ &- \sum_{Element} \int_0^l \left[f\delta u - q_y\delta v - q_z\delta w + r\delta\theta\right] ds - \sum_{Joint,q} \boldsymbol{Q}_J \cdot \boldsymbol{\delta u}_J - \sum_{Joint,h} \boldsymbol{R}_J \cdot \boldsymbol{\delta u}_J = 0\end{aligned}$$

Assume:

$$\boldsymbol{\delta u}_J = \boldsymbol{0} \quad at\ Joint, h$$

We have the simplified form:

$$\begin{aligned}\delta\mathcal{L}_\epsilon(\boldsymbol{u}) = &\sum_{Element} \int \left[P\delta u' + M_y\delta v'' + M_z\delta w'' + T\delta\theta'\right] ds \\ &- \sum_{Element} \int_0^l \left[f\delta u - q_y\delta v - q_z\delta w + r\delta\theta\right] ds - \sum_{Joint,q} \boldsymbol{Q}_J \cdot \boldsymbol{\delta u}_J = 0\end{aligned}$$

Principle of Virtual Force

$$\begin{aligned}\delta V_c(\boldsymbol{P}) = &\sum_{Element} \int \left[u'\delta P + v''\delta M_y + w''\delta M_z + \theta'\delta T\right] ds \\ &= \sum_{Element} \int_0^l \left[u\delta P' - v\delta M_y'' - w\delta M_z'' + \theta\delta T'\right] ds + \sum_{Joint,q} \boldsymbol{u}_J \cdot \boldsymbol{\delta Q}_J + \sum_{Joint,h} \boldsymbol{H}_J \cdot \boldsymbol{\delta Q}_J\end{aligned}$$

$$\delta W_{c,ext}(\boldsymbol{P}) = \sum_{Element} \int_0^l \left[u\delta P' - v\delta M_y'' - w\delta M_z'' + \theta\delta T'\right] ds + \sum_{Joint,q} \boldsymbol{u}_J \cdot \boldsymbol{\delta Q}_J + \sum_{Joint,h} \boldsymbol{H}_J \cdot \boldsymbol{\delta Q}_J$$

$$W_{c,ext}(\boldsymbol{P}) = \sum_{Element} \int_0^l \left\{u[P' + f] - v(\boldsymbol{P})\left[M_y'' + q_y\right] - w(\boldsymbol{P})[M_z'' + q_z] + \theta(\boldsymbol{P})[T' + r]\right\} ds$$
$$+ \sum_{Joint,q} \boldsymbol{u}_J \cdot \boldsymbol{\delta Q}_J + \sum_{Joint,h} \boldsymbol{H}_J \cdot \boldsymbol{\delta Q}_J$$

Please note that $\boldsymbol{u}$ is the generalized displacement field, $\boldsymbol{P}$ is the generalized force field and $\boldsymbol{Q}_J$ is the generalized nodal force:

$$\boldsymbol{u} = (Global): U\boldsymbol{E}_1 + V\boldsymbol{E}_2 + W\boldsymbol{E}_3 + \Theta\boldsymbol{E}_4 + \mathrm{A}\boldsymbol{E}_5 + \mathrm{B}\boldsymbol{E}_6$$
$$= (Local): u\boldsymbol{e}_1 + v\boldsymbol{e}_2 + w\boldsymbol{e}_3 + \theta\boldsymbol{e}_4 + \alpha\boldsymbol{e}_5 + \beta\boldsymbol{e}_6$$

$$\boldsymbol{P}, \boldsymbol{Q}_J(Global): F_X\boldsymbol{E}_1 + F_Y\boldsymbol{E}_2 + F_Z\boldsymbol{E}_3 + M_X\boldsymbol{E}_4 + M_Y\boldsymbol{E}_5 + M_Z\boldsymbol{E}_6$$
$$= (Local): P\boldsymbol{e}_1 - V_y\boldsymbol{e}_2 - V_z\boldsymbol{e}_3 + T\boldsymbol{e}_4 - M_z\boldsymbol{e}_5 + M_y\boldsymbol{e}_6$$

Lagrangian:

$$\mathcal{L}_c(\boldsymbol{u}) = V_c(\boldsymbol{u}) - W_{c,ext}(\boldsymbol{u})$$

$$\delta\mathcal{L}_c(\boldsymbol{P}) = \sum_{Element} \int \left[u'\delta P + v''\delta M_y + w''\delta M_z + \theta'\delta T\right] ds$$
$$- \sum_{Element} \int_0^l \left[u\delta P' - v\delta M_y'' - w\delta M_z'' + \theta\delta T'\right] ds - \sum_{Joint,q} \boldsymbol{u}_J \cdot \boldsymbol{\delta Q}_J - \sum_{Joint,h} \boldsymbol{H}_J \cdot \boldsymbol{\delta Q}_J$$
$$= 0$$

Assume:

$$\delta P'(\boldsymbol{P}) = \delta M_y''(\boldsymbol{P}) = \delta M_z''(\boldsymbol{P}) = \delta T'(\boldsymbol{P}) = 0 \quad on\ Element$$

$$\boldsymbol{\delta Q}_J = \boldsymbol{0} \quad at\ Joint, q$$

We have the simplified form:

$$\delta\mathcal{L}_c(\boldsymbol{u}) = \sum_{Element} \int \left[u'\delta P + v''\delta M_y + w''\delta M_z + \theta'\delta T\right] ds - \sum_{Joint,h} \boldsymbol{H}_J \cdot \boldsymbol{\delta Q}_J = 0$$

Castigliano's 1st Theorem

$$\delta\mathcal{L}_\epsilon(\boldsymbol{u}) = \sum_{Element} \int \left[P\delta u' + M_y\delta v'' + M_z\delta w'' + T\delta\theta'\right] ds$$
$$- \sum_{Element} \int_0^l \left[f\delta u - q_y\delta v - q_z\delta w + r\delta\theta\right] ds - \sum_{Joint,q} \boldsymbol{Q}_J \cdot \boldsymbol{\delta u}_J = 0$$

Castigliano's first theorem states that, if there is a variational generalized displacement field $\boldsymbol{\delta u}$ that makes $\boldsymbol{\delta u}_J = \delta\mathcal{C}\boldsymbol{u}_\Lambda$ at joint Λ, then the force at joint:

$$\delta\mathcal{C}\left\{\boldsymbol{Q}_\Lambda \cdot \boldsymbol{u}_\Lambda = \sum_{Element}\int\left[Pu'(\boldsymbol{u}_\Lambda) + M_y v''(\boldsymbol{u}_\Lambda) + M_z w''(\boldsymbol{u}_\Lambda) + T\theta'(\boldsymbol{u}_\Lambda)\right]ds \right.$$
$$- \sum_{Element}\int_0^l\left[fu(\boldsymbol{u}_\Lambda) - q_y v(\boldsymbol{u}_\Lambda) - q_z w(\boldsymbol{u}_\Lambda) + r\theta(\boldsymbol{u}_\Lambda)\right]ds$$
$$\left. - \sum_{Joint,q}\boldsymbol{Q}_J \cdot \boldsymbol{u}_J(\boldsymbol{u}_\Lambda)\ \ (except\ joint\ \Lambda)\right\}$$

$$\boldsymbol{Q}_\Lambda \cdot \boldsymbol{u}_\Lambda = \sum_{Element}\int\left[Pu'(\boldsymbol{u}_\Lambda) + M_y v''(\boldsymbol{u}_\Lambda) + M_z w''(\boldsymbol{u}_\Lambda) + T\theta'(\boldsymbol{u}_\Lambda)\right]ds$$
$$- \sum_{Element}\int_0^l\left[fu(\boldsymbol{u}_\Lambda) - q_y v(\boldsymbol{u}_\Lambda) - q_z w(\boldsymbol{u}_\Lambda) + r\theta(\boldsymbol{u}_\Lambda)\right]ds$$
$$- \sum_{Joint,q}\boldsymbol{Q}_J \cdot \boldsymbol{u}_J(\boldsymbol{u}_\Lambda)\ \ (except\ joint\ \Lambda)$$

With the partition of regions introduced in the previous chapter, we can separate the element at any point so that its internal forces become external boundary forces.

The first theorem in most cases cannot be used to obtain the forces directly, because the terms (P, M_y, M_z, T), $[u(\boldsymbol{u}_\Lambda), v(\boldsymbol{u}_\Lambda), w(\boldsymbol{u}_\Lambda), \theta(\boldsymbol{u}_\Lambda)]$, $[u'(\boldsymbol{u}_\Lambda), v''(\boldsymbol{u}_\Lambda), w''(\boldsymbol{u}_\Lambda), \theta'(\boldsymbol{u}_\Lambda)]$ are generally the result at very end instead of the known variables for a statically indeterminate structure. However, it provides an invaluable tool to discretize the linear structural system into sub-elements with supplemental boundary conditions.

If we could have $\delta u' = \delta v'' = \delta w'' = \delta\theta' = 0$ (This assumption can only be achieved in a statically determinate element since it means no deformation is generated under the variational displacement, which automatically grant $\boldsymbol{u}_J(\boldsymbol{u}_\Lambda) = \boldsymbol{0}$ for all other joints):

$$\boldsymbol{Q}_\Lambda \cdot \boldsymbol{u}_\Lambda = -\sum_{Element}\int_0^l\left[fu(\boldsymbol{u}_\Lambda) - q_y v(\boldsymbol{u}_\Lambda) - q_z w(\boldsymbol{u}_\Lambda) + r\theta(\boldsymbol{u}_\Lambda)\right]ds$$

Castigliano's 2nd Theorem (Space Frame)

$$\delta\mathcal{L}_c(\boldsymbol{u}) = \sum_{Element}\int\left[u'\delta P + v''\delta M_y + w''\delta M_z + \theta'\delta T\right]ds - \sum_{Joint,h}\boldsymbol{H}_J \cdot \boldsymbol{\delta Q}_J = 0$$

With the partition of regions introduced in the previous chapter, we can separate the element at any point so that its internal displacement becomes external boundary displacement.

Castigliano's second theorem states that if there is a variational generalized force field $\boldsymbol{\delta P}$ that makes $\boldsymbol{\delta Q}_J = \delta\mathcal{C}\boldsymbol{Q}_\Lambda$ at joint Λ, then the displacement at joint:

$$\delta\mathcal{C}\left\{\boldsymbol{H}_\Lambda \cdot \boldsymbol{Q}_\Lambda = \sum_{Element} \int [u'P(\boldsymbol{Q}_\Lambda) + v''^{M_y}(\boldsymbol{Q}_\Lambda) + w''M_z(\boldsymbol{Q}_\Lambda) + \theta'T(\boldsymbol{Q}_\Lambda)]\, ds - \sum_{Joint,h} \boldsymbol{H}_J \cdot \boldsymbol{Q}_J(\boldsymbol{Q}_\Lambda)\ \ (except\ joint\ \Lambda)\right\}$$

$$\boldsymbol{H}_\Lambda \cdot \boldsymbol{Q}_\Lambda = \sum_{Element} \int \left[u'P(\boldsymbol{Q}_\Lambda) + v''M_y(\boldsymbol{Q}_\Lambda) + w''M_z(\boldsymbol{Q}_\Lambda) + \theta'T(\boldsymbol{Q}_\Lambda)\right] ds - \sum_{Joint,h} \boldsymbol{H}_J \cdot \boldsymbol{Q}_J(\boldsymbol{Q}_\Lambda)\ \ (except\ joint\ \Lambda)$$

This statement seems too good to be true. In fact, certain conditions need to be met to utilize this simplified equation. Recalling the assumption of simplification, the variational generalized force field $\boldsymbol{\delta P} = \delta\mathcal{C}\boldsymbol{P}(\boldsymbol{Q}_\Lambda)$ must satisfy the homogeneous equilibrium on the elements:

$$\delta P'(\boldsymbol{P}) = \delta M_y''(\boldsymbol{P}) = \delta M_z''(\boldsymbol{P}) = \delta T'(\boldsymbol{P}) = 0 \quad on\ Element$$

Plus, the homogeneous equilibrium at joints:

$$\boldsymbol{\delta Q}_J = \boldsymbol{0} \quad at\ Joint, q$$

This is equivalent to having the reaction of the structure under a unit force at point.